Building Equivalent Ratio Tables

Grade 6–7 Mathematics

Completing ratio tables, finding the scale factor between rows, adding a unit-rate column, and deciding whether two tables are equivalent. Work with the same habit every time:

- A ratio table stays equivalent when you multiply (or divide) *both* entries of a column by the same number.
- Find that number first — "what times the old entry gives the new one?" — then apply it to the partner entry.
- Adding the same amount to both entries does **not** keep a ratio equivalent. $2 : 5$ and $4 : 7$ are different ratios.
- The unit rate is the column whose first entry is 1; it is the fastest way to compare two tables.

1. A muffin recipe uses 2 cups of flour for every 5 cups of milk. Complete the ratio table.

Flour (cups)	2	4	6	10
Milk (cups)	5			

Answer: ________________

2. One row of a ratio table reads 3 red beads to 8 white beads. Another row of the same table has 21 red beads. By what number was the first row multiplied, and how many white beads are in that row?

Red beads	3	21
White beads	8	

Answer: ________________

3. A school keeps a ratio of 5 teachers for every 90 students. Complete the table by scaling from the *student* row.

Teachers	5		
Students	90	180	270

Answer: ________________

4. A bottling machine fills 6 bottles every 4 seconds. Add the unit-rate column to the table: how many bottles does it fill in 1 second?

Bottles	6	
Seconds	4	1

Answer: ________________

5. Ana's table pairs 4 green tiles with 6 blue tiles. Ben's table pairs 14 green tiles with 21 blue tiles. Are the two tables equivalent? Write the two ratios as fractions in lowest terms and join them with $<$, $>$, or $=$.

Answer: ________________

6. A ratio table scales 8 minutes of video to 20 megabytes of storage. The next column shows 12 minutes. Find the scale factor from the first column to the second, then find the missing storage entry.

Minutes	8	12
Megabytes	20	

Answer: ________________

7. A paint mix uses 3 parts blue to 5 parts yellow. Fill in both missing entries. Notice that one column scales *up* from the first column and the other does not use a whole-number scale factor.

Blue (mL)	3		15
Yellow (mL)	5	20	

Answer: ________________

8. A train covers 150 miles in 4 hours at a steady speed. Add the unit-rate column to the table, then state the speed.

Miles	150	
Hours	4	1

Answer: ________________ mph

9. Two lemonade recipes are given as ratio tables of scoops of mix to cups of water. Recipe P uses 3 scoops to 5 cups; recipe Q uses 5 scoops to 8 cups. Which recipe tastes stronger, that is, which has more mix per cup of water? Compare the two ratios with $<$, $>$, or $=$ and say which recipe wins.

Answer: ________________

10. A ratio table is built from 5 : 3. One column of the table reads x to 42. Write and solve an equation to find x.

Answer: ________________

11. A ratio table starts with the column 9 to 12. A later column of the same table has 84 in the second row. Let k be the scale factor. Write and solve an equation for k, then find the missing first-row entry.

Answer: ________________

12. **Synthesis challenge.** A paint mix keeps blue to yellow in the ratio 4 : 7. One batch uses 66 ounces of paint in total. Build a ratio table with a "total" row, find how many equal parts the 66 ounces splits into, and state how many ounces of blue and how many ounces of yellow the batch contains.

Answer: ______________